

02 - Technical Operator Runbook / ขั้นตอนปฏิบัติทางเทคนิคสำหรับ Admin

ScheduledDefrag Trigger Validation + Cross-Layer Hang/Delay RCA

Windows Server 2022 + Veritas + VMware + HPE Alletra 9060

Important Concept / แนวคิดสำคัญ

English

Steps 4-16 are the **standard execution block for one scenario**.

You must repeat Steps 4-16 for each scenario:

```
Scenario A: C:\Temp\TrimTest-A-Baseline
Scenario B: C:\Temp\TrimTest-B-DefragDisabled
Scenario C: C:\Temp\TrimTest-C-Mitigation
```

Each scenario must use a separate OutputDir.

Each scenario requires a large SQL backup file to exist before deletion.

If the same file is used, recreate or copy the file again before starting the next scenario.

Main-plan note:

```
Scenario A/B/C do not include a manual Optimize-Volume / ReTrim command.
If manual optimizer confirmation is needed, run it only as an optional
appendix test after the main A/B/C plan is complete.
```

ภาษาไทย

Step 4-16 คือ **standard execution block** สำหรับ 1 scenario

ต้องทำซ้ำ Step 4-16 สำหรับแต่ละ scenario:

```
Scenario A: C:\Temp\TrimTest-A-Baseline
Scenario B: C:\Temp\TrimTest-B-DefragDisabled
```

Scenario C: C:\Temp\TrimTest-C-Mitigation

แต่ละ scenario ต้องใช้ OutputDir แยกกัน เพื่อไม่ให้ evidence ปนกัน

แต่ละ scenario ต้องมีไฟล์ SQL backup ขนาดใหญ่ให้ลบก่อนเริ่มทดสอบ

ถ้าใช้ไฟล์เดิม ต้อง copy/recreate ไฟล์กลับมาก่อนเริ่ม scenario ถัดไป

Common Preparation / เตรียมก่อนเริ่มทดสอบ

Step 1 - Prepare Script Folder / เตรียม Folder สำหรับ Script

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
mkdir C:\Temp\TrimTest-Scripts  
cd C:\Temp\TrimTest-Scripts
```

Copy these scripts into C:\Temp\TrimTest-Scripts:

```
03-TrimLatencyTrace.ps1  
04-Start-PerfMonBLG.ps1  
05-Stop-PerfMonBLG.ps1  
06-Collect-StorageEventSnapshot.ps1  
07-Collect-VeritasWindowsEvidence.ps1  
08-Export-WindowsTrimEvidence.ps1
```

Expected:

```
Scripts are available in C:\Temp\TrimTest-Scripts
```

Step 2 - Prepare External Collector Workstations / เตรียมเครื่องมือภายนอก

Where:

VMware PowerCLI workstation
HPE Alletra 9060 management workstation

Estimated:

10 minutes

Run / Prepare:

VMware workstation:
09-Collect-VMwareLatency.ps1

HPE Alletra 9060 management workstation:
10-Collect-3PARLatency.ps1

Expected:

VMware PowerCLI access confirmed
HPE Alletra 9060 SSH access confirmed

Step 3 - Prepare Customer Test File Prerequisites / เตรียมความพร้อมของไฟล์ทดสอบจากฝั่งลูกค้า

Where:

Customer application / backup owner
Windows Server 2022

Estimated:

This should be completed before the test window starts.

Confirm:

Item	What must be confirmed
File path	Full path of the large SQL backup file

Item	What must be confirmed
File size	Target size, ideally around 1.2 TB or agreed equivalent
File type	Example: <code>.bak</code>
Target volume	Drive letter / volume that will be tested
Delete approval	Customer confirms the file can be safely deleted
Recreate method	Copy back / restore / regenerate method for next scenario
Recreate owner	Person/team responsible for preparing the file again
Recreate duration	Approximate time needed before the next scenario

Expected:

The customer-side file preparation plan is confirmed before execution day.

Step 4 - Confirm Target File and Recreate Plan / ยืนยันไฟล์ที่จะลบ

Where:

Windows Server 2022

Estimated:

5 minutes

Run / Confirm:

Confirm large SQL backup file path.
Confirm file can be safely deleted.
Confirm how the file will be recreated/copied for next scenario.

Example file path:

D:\Backup\LargeSqlBackup.bak

Expected:

Target file path confirmed.
File recreation plan confirmed.

Scenario Configuration Matrix / ตารางตั้งค่า Scenario

Use the configuration below before running the standard execution block.

Scenario	OutputDir	DisableDeleteNotify	ScheduledDefrag
A - Baseline	C:\Temp\TrimTest-A-Baseline	0	Enabled/current
B - Defrag Disabled	C:\Temp\TrimTest-B-DefragDisabled	0	Disabled
C - Mitigation	C:\Temp\TrimTest-C-Mitigation	1	Disabled

Scenario A - Baseline / Reproduce

Configure Scenario A

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

5 minutes

Run:

```
$OutputDir = "C:\Temp\TrimTest-A-Baseline"

fsutil behavior set DisableDeleteNotify 0
fsutil behavior query DisableDeleteNotify

Enable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName "ScheduledDefrag"

Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName "ScheduledDefrag" |
Select-Object TaskName, State
```

```
Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
NTFS DisableDeleteNotify = 0  
ScheduledDefrag enabled/current  
No unexpected defrag.exe before start
```

Then run:

```
Step 5-17 using OutputDir = C:\Temp\TrimTest-A-Baseline
```

Scenario B - Trigger Isolation / ScheduledDefrag Disabled

Configure Scenario B

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-B-DefragDisabled"  
  
fsutil behavior set DisableDeleteNotify 0  
fsutil behavior query DisableDeleteNotify  
  
Disable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName  
"ScheduledDefrag"  
  
Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName  
"ScheduledDefrag" |  
Select-Object TaskName, State
```

```
Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
NTFS DisableDeleteNotify = 0  
ScheduledDefrag disabled  
No defrag.exe running
```

Then run:

```
Step 5-17 using OutputDir = C:\Temp\TrimTest-B-DefragDisabled
```

Scenario C - Mitigation Validation / Known-Good Control

Configure Scenario C

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-C-Mitigation"

fsutil behavior set DisableDeleteNotify 1
fsutil behavior query DisableDeleteNotify

Disable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
"ScheduledDefrag"

Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
"ScheduledDefrag" |
Select-Object TaskName, State

Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
NTFS DisableDeleteNotify = 1
ScheduledDefrag disabled
No defrag.exe running
```

Then run:

```
Step 5-17 using OutputDir = C:\Temp\TrimTest-C-Mitigation
```

Standard Execution Block / ขั้นตอนมาตรฐานสำหรับ 1 Scenario

Step 5 - Create Scenario Output Folder / สร้าง Output Folder

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
1 minute
```

Run:


```
New-Item -ItemType Directory -Path $OutputDir -Force | Out-Null
```

Expected:

```
Scenario OutputDir created.
```

Step 6 - Collect Pre-check Storage Snapshot / เก็บ Storage Snapshot ก่อนทดสอบ

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts  
.\06-Collect-StorageEventSnapshot.ps1 -OutputDir "$OutputDir\PreCheck"
```

Expected:

```
storage-snapshot-*.zip created under <OutputDir>\PreCheck
```

Step 7 - Collect Baseline Veritas Evidence / เก็บ Veritas Baseline

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

10 minutes

Run:

```
cd C:\Temp\TrimTest-Scripts
.\07-Collect-VeritasWindowsEvidence.ps1 -OutputDir "$OutputDir\Veritas"
```

Expected:

```
veritas-windows-*.zip created under <OutputDir>\Veritas
```

Step 8 - Start Windows Collectors / เริ่ม Windows Collectors

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

5 minutes

Run:

```
cd C:\Temp\TrimTest-Scripts

.\03-TrimLatencyTrace.ps1 -Action Start -OutputDir $OutputDir -
SampleInterval 5

.\04-Start-PerfMonBLG.ps1 -OutputDir $OutputDir -SampleInterval 5
```

Verify:

Get-Job

Expected:

```
TrimTest-Counters      Running
TrimTest-DefragProcess Running
TrimTest-DefragEvents  Running
TrimTest-Fsutil        Running
```

```
disk_latency_perfmon.blg created under <OutputDir>
```

Step 9 - Start VMware Collector / เริ่ม VMware Collector

Where:

```
VMware PowerCLI workstation
```

Estimated:

```
5 minutes
```

Run:

```
.\09-Collect-VMwareLatency.ps1 `
  -vCenter "vcenter.example.com" `
  -VMName "SQL-Windows-VM" `
  -DatastoreName "DATASTORE_NAME" `
  -OutputDir "<ScenarioOutputDir>\VMware" `
  -SampleInterval 20
```

Example:

```
.\09-Collect-VMwareLatency.ps1 `
  -vCenter "vcenter.example.com" `
  -VMName "SQL-Windows-VM" `
  -DatastoreName "DATASTORE_NAME" `
  -OutputDir "C:\Temp\TrimTest-B-DefragDisabled\VMware" `
  -SampleInterval 20
```

Expected:

```
vmware-vm-disk-latency.csv
vmware-datastore-latency.csv
```

Important:

Keep this collector running until Step 15.

Step 10 - Start HPE Alletra 9060 Collector / เริ่ม HPE Alletra 9060 Collector

Where:

Management workstation with SSH access to HPE Alletra 9060

Estimated:

5 minutes

Run:

```
.\10-Collect-3PARLatency.ps1 `
  -Array "3par-array.example.com" `
  -User "monitoruser" `
  -HostFilter "SQL_HOST_NAME" `
  -OutputDir "<ScenarioOutputDir>\3PAR" `
  -SampleInterval 5
```

Example:

```
.\10-Collect-3PARLatency.ps1 `
  -Array "3par-array.example.com" `
  -User "monitoruser" `
  -HostFilter "SQL_HOST_NAME" `
  -OutputDir "C:\Temp\TrimTest-B-DefragDisabled\3PAR" `
  -SampleInterval 5
```

Expected:

3par-latency-*.log
3par-commands-*.txt

Important:

Keep this collector running until Step 15.

Step 11 - Mark Before Delete / Mark ก่อนลบไฟล์

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

1 minute

Run:

```
cd C:\Temp\TrimTest-Scripts  
  
.\03-TrimLatencyTrace.ps1 -Action Mark -OutputDir $OutputDir -Message  
"Before delete SQL backup"
```

Expected:

timeline_markers.csv updated.

Step 12 - Delete Large SQL Backup File / ลบไฟล์ SQL Backup ขนาดใหญ่

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

1–5 minutes

Run:

```
Remove-Item "D:\Backup\LargeSqlBackup.bak" -Force
```

Expected:

```
Large SQL backup file deleted.
```

Important:

```
Replace D:\Backup\LargeSqlBackup.bak with the actual file path.  
Before running the next scenario, recreate/copy this file again.
```

Step 13 - Mark After Delete / Mark หลังลบไฟล์

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
1 minute
```

Run:

```
cd C:\Temp\TrimTest-Scripts  
  
.\03-TrimLatencyTrace.ps1 -Action Mark -OutputDir $OutputDir -Message  
"After delete SQL backup"
```

Expected:

```
timeline_markers.csv updated.
```

Step 14 - Observe 60+ Minutes / Monitor อย่างน้อย 60 นาที

Where:

```
Windows Server 2022  
VMware PowerCLI workstation  
HPE Alletra 9060 management workstation
```

Estimated:

```
60 minutes minimum
```

Optional Windows status check:

```
cd C:\Temp\TrimTest-Scripts  
  
.\03-TrimLatencyTrace.ps1 -Action Status -OutputDir $OutputDir
```

Watch for:

```
hang/slowness  
defrag.exe activity  
Defrag Operational events  
Windows disk latency spike  
Windows disk queue spike  
VMware datastore latency spike  
HPE Alletra 9060 latency spike  
Veritas / path / disk events
```

If hang/slowness occurs:

```
cd C:\Temp\TrimTest-Scripts  
  
.\06-Collect-StorageEventSnapshot.ps1 -OutputDir "$OutputDir\DuringSpike"  
  
.\07-Collect-VeritasWindowsEvidence.ps1 -OutputDir "$OutputDir\Veritas"
```

Expected:

```
Observation completed with or without issue.  
If issue occurs, DuringSpike evidence captured.
```

Step 15 - Stop Windows Collectors / หยุด Windows Collectors

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts  
.<\03-TrimLatencyTrace.ps1 -Action Stop -OutputDir $OutputDir  
.<\05-Stop-PerfMonBLG.ps1
```

Expected:

```
Windows collector jobs stopped.  
PerfMon BLG stopped.
```

Step 16 - Stop VMware and HPE Alletra 9060 Collectors / หยุด VMware และ HPE Alletra 9060

Where:

```
VMware PowerCLI workstation  
HPE Alletra 9060 management workstation
```

Estimated:

```
1 minute
```

Run:

```
Press Ctrl+C in each collector window.
```


Expected:

```
VMware CSV collection stopped.  
HPE Alletra 9060 log collection stopped.
```

Step 17 - Export Final Evidence / Export Evidence หลังจบ Scenario

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
10–15 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts  
  
.\08-Export-WindowsTrimEvidence.ps1 -OutputDir $OutputDir  
  
.\06-Collect-StorageEventSnapshot.ps1 -OutputDir "$OutputDir\PostTest"  
  
.\07-Collect-VeritasWindowsEvidence.ps1 -OutputDir "$OutputDir\Veritas"
```

Expected:

```
windows-eventlogs-*.zip created  
storage-snapshot-*.zip created  
veritas-windows-*.zip created
```

After Each Scenario / หลังจบแต่ละ Scenario

Before moving to the next scenario:

1. Confirm all collectors are stopped.
2. Confirm evidence files exist in the scenario OutputDir.
3. Recreate/copy the large SQL backup file again.

4. Configure the next scenario.
5. Repeat Step 5–17.

Final Evidence Checklist / Checklist หลักฐานสุดท้าย

After all scenarios, evidence should be separated as:

```
C:\Temp\TrimTest-A-Baseline  
C:\Temp\TrimTest-B-DefragDisabled  
C:\Temp\TrimTest-C-Mitigation
```

Each folder should contain:

```
disk_latency_counters.csv  
disk_latency_perfmon.blg  
defrag_process_monitor.csv  
defrag_event_monitor.csv  
fsutil_trim_status.csv  
timeline_markers.csv  
environment_snapshot.txt  
summary_after_stop.txt  
windows-eventlogs-*.zip  
PreCheck\storage-snapshot-*.zip  
PostTest\storage-snapshot-*.zip  
Veritas\veritas-windows-*.zip  
VMware\vmware-vm-disk-latency.csv  
VMware\vmware-datastore-latency.csv  
3PAR\3par-latency-*.log  
3PAR\3par-commands-*.txt
```

Appendix A - Optional Manual Optimize-Volume / ReTrim Confirmation

Use this appendix only if the customer requests extra confirmation after Scenario A/B/C are complete.

Purpose:

```
Confirm whether a manually triggered optimizer/retrim pass reproduces the  
same delay pattern  
after the file deletion has already completed.
```

Important:

This appendix is optional and is not part of the main customer A/B/C plan. Use a separate OutputDir so the optional evidence does not mix with the main evidence set.

Recommended OutputDir:

```
C:\Temp\TrimTest-D-ManualRetrim
```

Appendix A Step 1 - Configure Optional Scenario

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-D-ManualRetrim"

fsutil behavior set DisableDeleteNotify 0
fsutil behavior query DisableDeleteNotify

Disable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
"ScheduledDefrag"

Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
"ScheduledDefrag" |
Select-Object TaskName, State

Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
DisableDeleteNotify = 0
ScheduledDefrag disabled
No unexpected defrag.exe before start
```

Appendix A Step 2 - Run Standard Execution Block Through Step 12

Run Step 5-13 using:

```
OutputDir = C:\Temp\TrimTest-D-ManualRetrim
```

Important:

Do not stop the collectors after Step 13.

Appendix A Step 3 - Mark Before Manual ReTrim

Where:

Windows Server 2022 PowerShell – Run as Administrator

Run:

```
cd C:\Temp\TrimTest-Scripts  
  
.\03-TrimLatencyTrace.ps1 -Action Mark -OutputDir $OutputDir -Message  
"Before manual Optimize-Volume ReTrim"
```

Appendix A Step 4 - Run Manual Optimize-Volume / ReTrim

Where:

Windows Server 2022 PowerShell – Run as Administrator

Run:

```
Optimize-Volume -DriveLetter D -ReTrim -Verbose
```

Important:

Replace drive letter D with the actual target volume drive letter.
Run the command only after evidence collectors are already running.

Appendix A Step 5 - Mark After Manual ReTrim

Run:

```
cd C:\Temp\TrimTest-Scripts

.\03-TrimLatencyTrace.ps1 -Action Mark -OutputDir $OutputDir -Message
"After manual Optimize-Volume ReTrim"
```

Appendix A Step 6 - Observe 60+ Minutes and Complete Step 13-16

Run:

Observe for at least 60 minutes, then complete Step 14-17.

Interpretation:

If the environment remains stable after file deletion but degrades only after manual ReTrim,
that strengthens the case for an optimizer-driven reclaim trigger.

Appendix B - Tuning Note

The following items are intentionally excluded from the main execution plan and from Appendix A execution unless separately approved for a later tuning phase:

```
vol_tp_reclaim_limit
vol_tp_reclaim_idle_delay
```